**Topology: MAT307/M405/MAT803 : - Problem Set** 6

1. Let $p : X \twoheadrightarrow Y$ be a closed continuous surjective map such that $p^{-1}(y)$ is compact for each $y \in Y$. Prove that if $Y$ is compact, then so is $X$.

2. Let $X$ be a compact metric space. Prove that if $f : X \to X$ is an isometry (distance preserving map) of $X$, then $f$ is a homeomorphism.

3. Prove that the composition of two quotient maps is a quotient map.

4. Prove that a surjective continuous map $f : X \twoheadrightarrow Y$ that is either an open map or a closed map is a quotient map.

5. Prove that a bijective quotient map is a homeomorphism.

6. Let $f_1 \colon X_1 \to Y_1$ and $f_2 \colon X_2 \to Y_2$ be maps between topological spaces, and let $f_1 \times f_2 \colon X_1 \times X_2 \to Y_1 \times Y_2$ denote the product map $(x_1, x_2) \mapsto (f_1(x_1), f_2(x_2))$.
   (a) Prove that if $f_1$ and $f_2$ are continuous, then $f_1 \times f_2$ is continuous with respect to the product topologies.
   (b) Give a counterexample to show that if $p_1 \colon X_1 \to Y_1$ and $p_2 \colon X_2 \to Y_2$ are quotient maps, $p_1 \times p_2$ need not be a quotient map.

7. Let $X$ and $Y$ be topological spaces, and $f : X \to Y$ be a bijective continuous map. Show that if $Y$ is Hausdorff, then $X$ is also Hausdorff.

8. Let $X$ be a compact topological space and $p : X \twoheadrightarrow Y$ be a quotient map.
   (a) Prove that $Y$ is compact.
   (b) Show that if $Y$ is Hausdorff, then $p$ is a closed map.

9. Let $p : X \to Y$ be a continuous surjection. Prove that $p$ is a closed map if and only if for every subset $S \subseteq Y$ and every open set $U \subseteq X$ containing $p^{-1}(S)$, there exists an open set $V \subseteq Y$ containing $S$ such that $p^{-1}(V) \subseteq U$.

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